Introduction

Units and measurement

Units and Measurement

1. Physical Quantities & Units
2. Apparatus / Instruments

APPENDIX C: PHYSICAL QUANTITIES AND UNITS

Fundamental physical quantities

Physical quantity	Unit(s)*
length	**metre (m)** kilometre (km) centimetre (cm) millimetre (m)
mass	**kilogram (kg)** gram (g) milligram (mg)
time	**second (s)** millisecond (ms)
temperature	**Kelvin (K)** degrees Celsius/°C
current	**ampere** or **amp (A)** milliampere or milliamp (mA)

*SI base units are shown in bold.

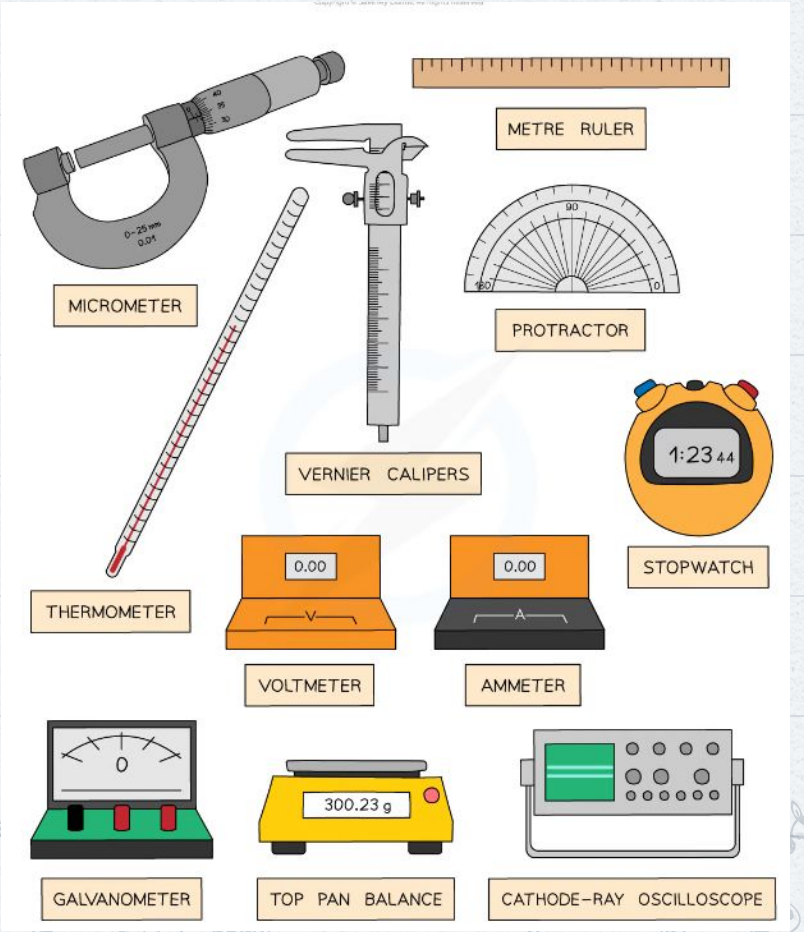
Derived quantities and units

Physical quantity	Unit(s)
acceleration	m/s^2
area	cm^2 m^2
density	kg/m^3 g/cm^3
electrical charge	coulomb (C)
electrical current	ampere (A)
energy (work)	joule (J) kilojoule (kJ) megajoule (MJ)
frequency	hertz (Hz) kilohertz (kHz)
force	newton (N)
gravitational field strength	N/kg
moment of a force	Nm Ncm
momentum	kg m/s
potential difference (voltage)	volt (V) millivolt (mV)
power	watt (W) kilowatt (kW) megawatt (MW)
pressure	pascal (Pa or N/m^2) N/cm^2
radioactivity	becquerel (Bq)
resistance	ohm (Ω)
speed/velocity	m/s km/h
volume	cm^3 m^3 litre (l) millilitre (ml)

Apparatus

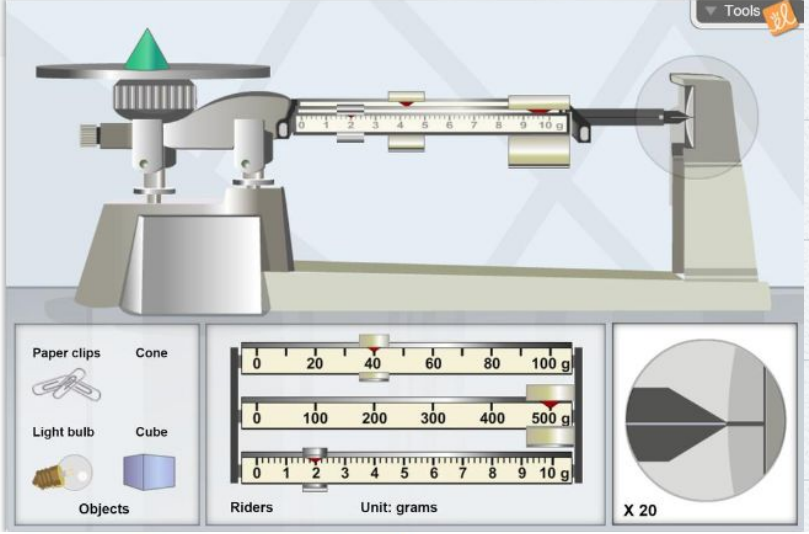
Base Physical Quantities		SI Unit	Measuring Apparatus (Alat ukur)
Time	Waktu	Second (s)	Stopwatch
Mass	Massa	Kilogram (kg)	Triple beam balance (neraca timbang)
Length	Panjang	Metre (m)	Vernier caliper, micrometer screw gauge (jangka sorong, mikrometer sekrup)
Temperature	Suhu	Kelvin (K)	Thermometer
Electric Current	Arus Listrik	Ampere (A)	Amperemeter
Luminous Intensity	Intensitas Cahaya	Candela (Cd)	Luxmeter
Amount of Substance	Jumlah Zat	Mol	

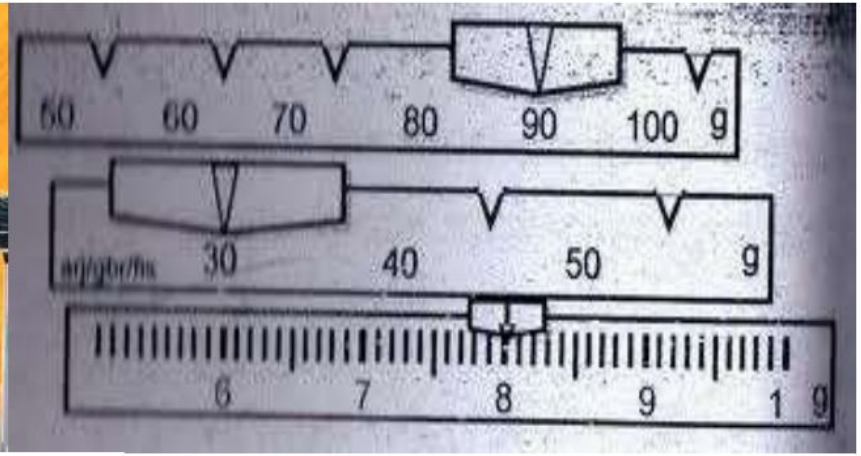
Apparatus



Apparatus	Purpose	Typical Resolution
Metre Rule	To measure the length of an object of moderate length	1 mm
Vernier Calipers	To measure short lengths	0.05 mm
Micrometer Screw Gauge	To measure small values of width, thickness, or diameter	0.001 mm
Top-pan Balance	To measure the mass of an object	0.01 g
Protractor	To measure angles	1°
Stopwatch	To measure periods of time	0.01 s
Thermometer	To measure the temperature of a body	1 °C
Voltmeter	To measure the potential difference across a component	1 mV – 0.1 V
Ammeter	To measure the electric current flowing through a component	1 mA – 0.1 A
Oscilloscope	To display waves and measure their frequencies	1 Hz
Laser	To provide a monochromatic source of light	450–650 nm

Triple Beam Balance

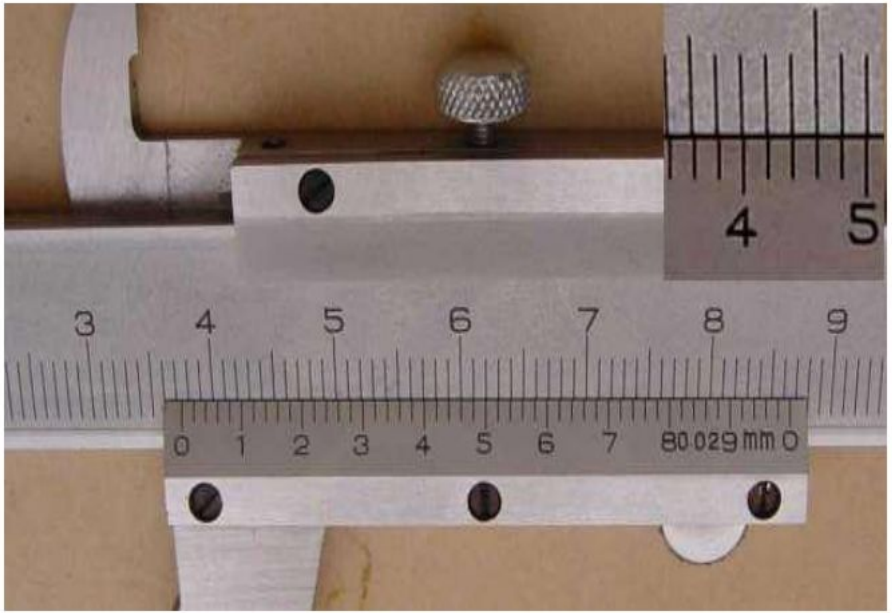




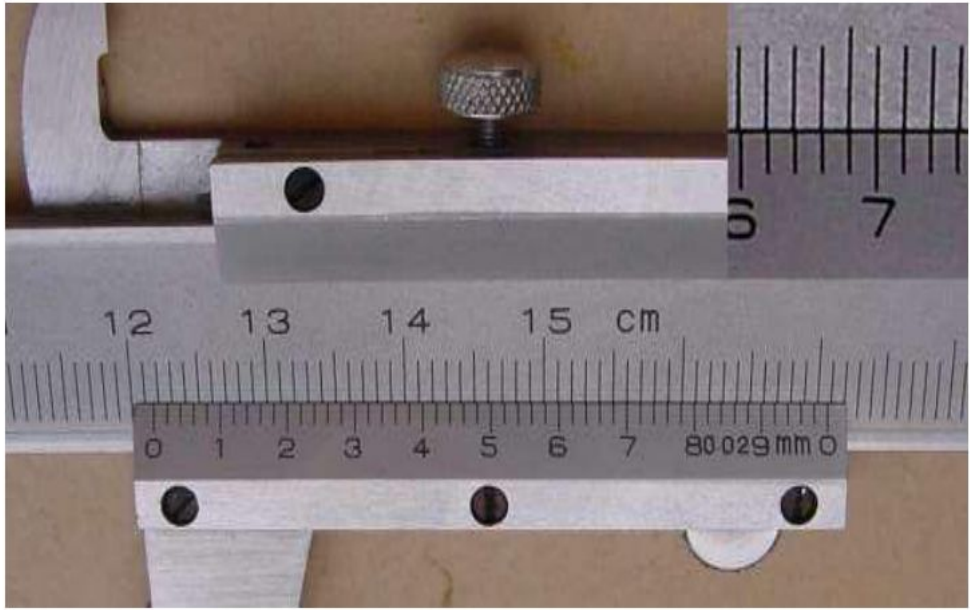
Triple beam balance; 168 g

128g

Vernier Caliper



Vernier caliper 3,746 cm



12,168 cm

Virtual Laboratory

Watch Video: https://youtu.be/pSaCqVQ7hZY

Try below virtual laboratory for several measuring tools :
Triple beam balance (do to get certificate of achievement)
http://www.thephysicsaviary.com/Physics/Programs/Games/ReadtheTripleBeam/index.html

Measuring/Graduated cylinder (do to get certificate of achievement)
http://www.thephysicsaviary.com/Physics/Programs/Games/GraduatedCylinder/index.html

Thermometer (do to get certificate of achievement)
http://www.thephysicsaviary.com/Physics/Programs/Games/ReadTheThermometerChallenge/index.html

Finding volume (do to get certificate of achievement)
http://cstephenmurray.com/onlinequizes/chemistry/measuring/displacementmethod.htm

Vernier caliper (try on how to measure using vernier caliper)
https://www.stefanelli.eng.br/en/virtual-vernier-caliper-milesimal-inch-50-simulator/